

Planting Survival and Success Study

Jefferson County Parks & Open Space (JCPOS) Small Research Grant Program 2025

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Executive Summary

In an effort to increase native plant applications and success in wildland and horticultural settings, we conducted a study in which we propagated and reared four forb and four grass species in a crossed design that measured plant growth and survival in 1) three types of soil (peat moss, a mineral and compost mix, and native garden soil), 2) with and without fertilization, and 3) with three planting dates (July 11, August 14, September 11). We coordinated three volunteer events to install the grown plant material in a restoration site on North Table Mountain Park. We coordinated biweekly ecological monitoring events to measure the success of these plants in the field. We engaged 114 volunteer community scientists to conduct monitoring protocols. University of Colorado - Boulder geography student interns and CoNPS members assisted in the report generation, data digitization, and analysis.

Our results indicate that plant species had variable survival rates compared to each other. Forb survival rates were higher than grass survival rates with the exception of *Bouteloua gracilis*, a top performer in the study. In all cross comparisons of soil, fertilization, and planting date, plants grown in the mineral compost mix (MC) had greater survival rates than plants grown in peat or garden soil (S3). Fertilized plants had greater survival rates than unfertilized plants. September-planted individuals had the highest survival rates at the end of the 2025 field season.

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The presence/absence survival measurement was not possible in the spring monitoring effort (April 2026). We could not ascertain individual plants due to the vegetation growth of some species. Other species had not yet emerged yet. Thirty five unique species, including six of seven planted species were observed in the plots. It was apparent that glyphosate application in the *Bromus inermis* dominated landscape provided niche space for other species to colonize the plots. Diversity increased due to the herbicide application, not simply due to our planting efforts.

This line of research comparing native plant growth success in peat and non-peat media has impressive implications for climate resiliency in today's world. No amount of gardening or restoration will restore the carbon sink that is lost when peat is harvested. If we start with this heavy burden of carbon sourcing, our restoration efforts will never achieve carbon neutrality, let alone carbon storage. This research is only a pilot study but hopefully it has set the course for JCPOS to take the most climate resilient pathways in its restoration pursuits.

Abstract

In the Jefferson County Parks & Open Space (JCPOS) native plant survival study, eight native species were grown and planted in a restoration site on North Table Mountain Park. The plants were grown in three soil types, including peat moss, a crushed granite (mineral) and compost mix, and native garden soil. Half of the plants were fertilized during the rearing phase. Plants were installed in the field in July, August, and September. Plants grown in the mineral and compost mix had the greatest survival

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rates, while peat moss-grown plants had the lowest survival rates across treatment levels. Native garden soil-grown plants had intermediate survival rates. Across all treatment levels, fertilized plants had the greatest survival rates. To date, plants installed in September have the highest survival rates. In our preliminary analysis of these data, all cross comparisons of treatment levels and plant survival indicate that native plants grown in mineral and compost mix have the greatest survival rates.

Literature Review

While there are significant literature contributions supporting native plant success in wildland restoration conditions, more work is needed to describe best management practices for high elevation revegetation using native plants, especially in landscapes facing intense recreational pressure and threats of wildfire and drought (Dreesen & Harrington, 1997; Gaddis, 2018; Meyer, 2006; Von Bargen, et al, 1988). We wanted to understand which native species are 1) representative of the local wild flora, and 2) easy to manage in smaller growing operations that might be implemented by municipal, county, and open space managers to facilitate local project needs, and 3) whether peat free soil media can be used in propagation efforts to reduce the total carbon footprint of growing operations. There are only a few nurseries in the western United States that can handle seed increase on a large scale and none in Colorado, other than the USFS Meeker Plant Materials Center, which can not possibly serve all the needs of Colorado communities and landscapes. Therefore, local land managers need to learn how to grow

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plants for their own restoration activities.

Given our tremendous success in the Pikes Peak Region with methods that provide *a priori* conclusions upon which this study is built, we were confident that our partnership with JCPOS would extend the impact of our work to increase native plants and habitats in Colorado, for the survival and well-being of humans and wildlife alike. In addition to our horticultural successes, we found that these methods greatly reduced the operational costs of the Colorado Springs (COS) city horticulture department, specifically the cost of soil and seed used in propagation efforts. We utilized more square footage in the horticultural complex while reducing electrical and heating costs by overwintering natives outdoors and in cold houses. We expect similar cost-saving results could be achieved by JCPOS.

Research Objective: Investigate native plant survival and growth rates across soil types, planting dates, and fertilization treatment levels.

Hypotheses

Hypothesis 1: Native seeds have variable germination rates that are regulated by environmental conditions.

Null Hypothesis 1: Environmental conditions do not affect seed germination rates.

Hypothesis 2: There is a difference in plant growth due to soil type.

Null Hypothesis 2: There is no difference in plant growth due to soil type.

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Hypothesis 4: There is a difference in plant growth with fertilization.

Null Hypothesis 4: There is no difference in plant growth with fertilization.

Hypothesis 5: There is a difference in plant growth due to planting date.

Null Hypothesis 5: There is no difference in plant growth due to planting date.

Methods

We planted over 2000 seeds of four native forb species and four native grass species (Table 1). All seed except the *Pascopyrum smithii* was hand collected on JCPOS properties. The *Pascopyrum smithii* came from a Boulder County seed increase. You can learn more about these species in this [slideshow](#). We established a crossed design with three soil treatments, three planting dates, and two fertilization regimes, fertilization and no fertilization. We worked with JCPOS to select the species from the JCPOS seed bank. All seeds used were native to JCPOS and collected by volunteers or staff on other JCPOS volunteer events prior to 2025.

Of note, we removed *Symphyotrichum ericoides* (SYER) because as it grew, we could see two leaf forms, and when it flowered, we could see three flower types, suggesting that this *Symphyotrichum* could have hybridized. We eventually keyed out a flowering specimen to *S. falcatum* and JCPOS staff keyed out other specimens to *S. laeve*. Interestingly, after removing this species from the study, lots of *Symphyotrichum* started appearing in our plots making it impossible to know which ones we planted.

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Table 1. Eight native species were used in this study.

Species Code Key
ACMI - <i>Achillea millefolium</i> common yarrow
ARLU - <i>Artemisia ludoviciana</i> white mountain sage/ silver wormwood
HEVI - <i>Heterotheca villosa</i> hairy goldenaster
SYER - <i>Symphyotrichum ericoides</i> white aster
BOGR - <i>Bouteloua gracilis</i> blue grama
HECO - <i>Hesperostipa comata</i> needle and thread grass
NAVI - <i>Nassella viridula</i> green needle
PASM - <i>Pascopyrum smithii</i> western wheatgrass

The first phase of this research occurred in the Colorado Springs city greenhouses. JCPOS supplied us with their chosen planting vessels, SC10s. We set up one SC10 tray per soil type per species. Each SC10 tray held 98 plants. We planted 98 seeds per forb species in two soil types for a total of 184 seeds per species and a total of 736 forb seeds total. We planted approximately 10 seeds in each grass cell to achieve a plug for planting. We planted the seed on 5-1-2025 and 5-2-2025.

These trays were watered twice per day with a mist spray that contained

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fertilizer. All the plants received fertilization during the germination phase because there was no way to standardize the watering in the greenhouse and also avoid fertilization.

Once the plants went outside for hardening off, the regime of fertilization or no fertilization began.

During the germination period, while the seedlings were in the COS city greenhouses, 12-6-6 natural liquid fertilizer was applied through the misting system at a rate of 16oz/500 gallons, which is a dilution rate of 0.03oz/gallon. During the hardening off period when the plants were outside, but not yet in the field site, we applied a 12-6-6 natural liquid fertilizer at a rate of 0.04oz/gallon, once per week, using a pump sprayer.

We measured germination rates weekly until June 1. We calculated germination rates on June 1, 2025. We then moved the plants outside to harden them off and began the fertilization regime. We fertilized half of the plants using a pump sprayer to mimic the foliar application applied in the greenhouse. The other half of the plants received only water.

Once the plants were outside, we began watering with an overhead, oscillating sprinkler every day for one week, then every other day for week two, then every three days for week three. We watered for 45 minutes each time. In week three, what had been a very wet season came to an abrupt end. This unfortunately collided with our water reduction schedule. We lost many plants in mid-June because the temperatures increased to 100 F and it suddenly stopped raining nearly every day.

We planted any additional plants from the germination effort in 3" pots and native

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garden soil. We never intended for these plants to enter the study environment, calling them the “insurance plan.” We intended to give them to JCPOS since the request was made to return all seed and plant material that was not used in the study design. When the originally germinated plants suffered during the hardening off process, we decided, in conversation with JCPOS staff, to install these plants in the native garden soil to maintain our total plant numbers for the study. This was an unfortunate outcome as it complicated the treatment levels. However, it was very interesting to observe the results discussed below. We stopped planting garden soil plants in August because we did not have enough prepared plots to carry out another round of planting three soil types.

We had a volunteer event at the Colorado Springs horticulture complex on July 1 during which we replanted a half set of seeds in SC10s for a total of 49 seeds per forb species, one seed per cell, per soil treatment. We planted approximately 10 seeds per cells, per soil treatment, for grass species, for an approximate total of 490 seeds per grass species planted. These plants were installed in the September 11 planting. As such, these plants were planted out at the same age as the original plant set, approximately two months after the seeding effort. Furthermore, they spent the same one month in the greenhouse and one month outside before being planted in the study plots.

The crossed design investigated an amendment treatment, 1) fertilizer, and 2) no fertilizer, in three soil types, 1) peat-based greenhouse potting soil commonly used in planting operations, 2) a proprietary mix of crushed granite (mineral) and compost made

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from Colorado Springs municipal horticultural waste, and 3) native garden soil. It also compared three planting dates 7/10/26, 8/14/26, 9/11/26.

The three soils employed were mineral-compost (MC), native garden soil (S3) and peat. We used Fertilome Ultimate, ready to use potting mix for the peat media. This is a Lambert product composed of 5-15% perlite and 85-95% sphagnum peat moss. One soil chemistry study has already been completed (Houghton, 2024), which indicated that the chemical composition of the mineral and compost mix was more like native soils than commercial nursery mix. Follow-up research indicated that native species were capable of germinating in all three soil types (Rose, 2024).

Garden soil (S3) was harvested from the ground in a residential context in Colorado Springs. At a previous time, subsoil had been bailed from another garden area and added to this site. This is the location from which we harvest soil to mimic the mineral-compost (MC) mix used at the COS greenhouses since it is heavy in mineral content. We mixed this with homemade compost composed primarily of chicken refuse, pine shavings, food scraps, and garden waste.

The MC soil is composed of a mineral product purchased at [C&C Sand and Stone](#), Colorado Springs, CO. The mineral base called “cimarron breeze” is mined from the Menzer quarry south of Colorado Springs. Cimarron breeze is composed of mineral particles $\frac{3}{8}$ ” and smaller. The compost is made at the city horticultural complex. It is made from horticultural waste, composed primarily of forestry waste materials, but also any waste from any annual flower beds. It contains some peat moss as the annual plant

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operations of the horticultural department are grown in peat moss, as they are purchased plants rather than grown on site from seeds.

We established 12 common garden plots on JCPOS property on North Table Mountain Park, with the guidance and assistance of JCPOS staff (Figure 1). To reduce the non-native plant density in the plots, JCPOS staff sprayed the 12 common garden plots with a 3% solution of Ranger Pro. Application took place 4/8/2025. On each planting day, we prepared the plots by handpulling any non-native species present, including encroaching *Bromus inermis*. We did not continue to weed the plots on subsequent monitoring visits.

> Glyphosate treatment 4/2025

> Planted with 7 native species

ACMI - *Achillea millefolium*
ARLU - *Artemisia ludoviciana*
BOGR - *Bouteloua gracilis*
HECO - *Hesperostipa comata*
HEVI - *Heterotheca villosa*
NAVI - *Nasella viridula*
PASM - *Pascopyrum smithii*

> 3 planting dates

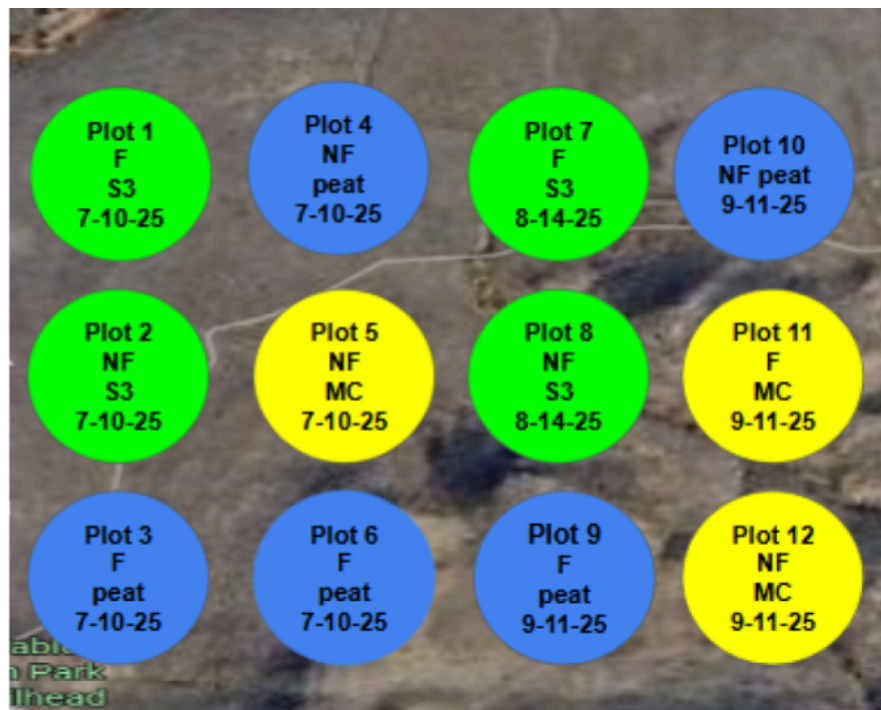
7-10-25, 8-14-25, 9-11-25

> 3 soil types, two pot sizes

- Peat, SC10
- MC, mineral-compost mix, SC10
- S3, garden soil, 3" pots

NF = not fertilized

F = fertilized



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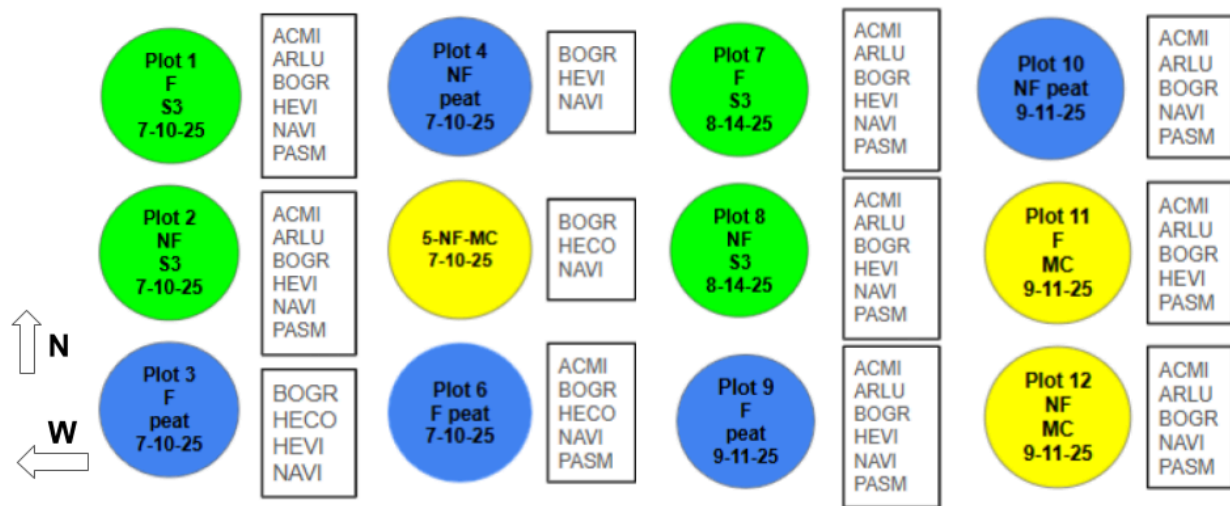


Figure 1. Schematic depicting plots with soil type, fertilization, and planting date indicated. ArcGIS online layer depicting this information is [hyperlinked here](#).

We planted out the first set of plants on July 10, 2025. We measured plant growth parameters (germination, plant height, number of stems) biweekly from July 2025 to October 2025 with volunteers recruited through the JCPOS Offero site and through the CoNPS website calendar. We calendared the events in Offero so volunteers could sign the JCPOS waiver. We then cross-linked those events into the CoNPS calendar. We planted out during three volunteer events (July 10, August 14, and September 11, 2025). On the July planting date, we planted all species in peat and S3. We also planted BOGR in MC, which was the only species with significant sample size that survived the hardening off stage. On the August planting date, we planted all species in MC, peat, and S3. On the September planting date, we planted all species in MC and peat. We did not plant any species in S3 because we had no more plots to separate the treatments.

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The volunteer demographics were a mix of JCPOS and CoNPS volunteers.

In biweekly monitoring events calendared in the same way described above for the planting dates, we measured plant growth parameters (presence, plant height, number of stems). We will conduct an additional monitoring event in the spring of 2026 to ascertain winter survival. We engaged 114 volunteers through our calendared events.

We also engaged four interns who helped run the events, complete the monitoring after volunteer events, digitize the datasheets, and analyze the data. Three interns were recruited from our CU Boulder Geography Department internship program. The two other interns were recruited through the volunteer events. While our data collection is complete and our preliminary data analysis is shared in this draft report, interns are still working to analyze these data using R software. Another intern will prepare a map of the study site. These are not completed yet because the interns are in a semester-long internship and their projects are not due until the end of the semester. We will incorporate these efforts into our final report.

We collected soil from each of the field plots for chemical analysis using the collection protocol offered by the [CSU Soil Testing Laboratory](#). We sent these samples to the CSU Soil Testing Laboratory along with a peat sample, MC, and S3 soil sample. We ordered a complete soil test and a texture analysis.

Soil pH primarily affects nutrient availability to plants. pH is a measure of H^+ ion activity in the soil solution. The more free H^+ ions there are, the more acidic the soil is. In acidic soils, excess H^+ competes with nutrient cations such as Mg^{2+} , K^+ , and Ca^{2+} for

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exchange sites on soil particles. When H^+ replaces these nutrients, the nutrients are released into the soil solution, where they may be leached away and become less available to plants (Whiting et al., 2002; Franzmeier et al., 2016). Colorado soils generally range from neutral to basic. Acidic soils are more typical in humid climates. In more basic soils, nutrients like zinc and iron are commonly in short supply (Whiting et al., 2002).

In this study, pH matters for two main reasons. Native plants evolve with their local soil chemistry. If seedlings are raised in a substrate that does not reflect the soil they evolved in (for example, acidic peat-based greenhouse mixes), they may experience stress when transplanted. If site pH is not within a species' tolerance range, seedling survival may decline. Fortunately, soil pH can be modified with amendments, at least temporarily, to support seedling establishment.

Soluble salts (Electrical Conductivity—EC) form naturally as water weathers rocks and minerals. Many salts are essential for plant function, but high concentrations can inhibit water uptake. In Colorado's arid climate, salts tend to accumulate because there is less rainfall to wash them deeper into the soil profile. Native plants are generally adapted to this, but salt sensitivity varies with growth stage. Seedlings are usually the least tolerant to excess soluble salts. (Morden et al., 1986).

Lowering salts in a restoration setting is challenging; the most effective method is leaching with water, which is not always feasible. For this study, knowing the EC (soluble salts) gives us a baseline and helps identify: 1) whether there is excess soluble salt at

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any of the plots which may affect seedling survival 2) which species are salt-tolerant and 3) whether planting should align with seasons of higher precipitation.

Organic matter plays a major role in determining soil structure. It acts like a “glue,” helping soil particles form aggregates. These aggregates regulate water movement, improve drainage in dense soils, and help sandy soils hold moisture. Organic matter also retains far more water and nutrients than mineral particles do. Soils low in OM tend to have lower fertility and dry out more quickly, factors that directly affect seedling survival in restoration contexts. (USDA, Role of organic matter).

The “lime index” (or buffer pH) reflects how resistant a soil is to a change in pH—its buffering capacity (Franzmeier et al., 2016).

Low lime index = soil will change pH easily.

High lime index = soil is more resistant to change.

This measurement is most often used in acidic soils where lime is added to raise the pH. For our study, lime indices help us understand how quickly pH-adjusting amendments would take effect, if needed.

CEC measures how many nutrients a soil can hold and exchange. Most soil particles have a negative charge, and many nutrient ions have a positive charge (K^+ , Ca^{2+} , Mg^{2+}). These nutrients temporarily attach to the soil surface and can be swapped with H^+ ions released by plant roots. The greater the negative charge and surface area of a soil (common in soils with high clay or organic matter), the greater its CEC

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(Franzmeier et al., 2016). In this study, higher CEC indicates soils that can hold more nutrients for longer. Because seedlings in all soil types respond positively to fertilization, CEC helps explain how much fertilizer the soil can retain before nutrients begin to leach away.

Nitrate (NO₃), Phosphorus (P), Potassium (K), Calcium (Ca) Magnesium (Mg), Sulfate-Sulfur(SO₄-S) are essential plant macronutrients. Limited amounts of any of these nutrients can restrict plant growth and functioning (Taiz et al., 2023).

Sodium (Na), Zinc (Zn), Iron (Fe), Manganese (Mn), Copper (Cu) are essential plant micronutrients: micro not meaning un-important. The micronutrients are just as essential to plant function, but the plants use lesser amounts of these nutrients (Taiz et al., 2023).

Results

Germination Results

We measured germination rate as presence/absence per cell. Since only one seed was planted per forb, this was the presence or absence of one seed. Since we planted approximately 10 seeds per cell of grass species, presence was met by any single individual germinated in the cell. Germination rates across all species in all soil types were impressive. This is firstly a testament to the JCPOS seed collection efforts as clearly the seed had been collected at an appropriate time of seed maturity. Germination rates were slightly higher in peat moss than in

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mineral compost mix, with the exception of *Hesperostipa comata* (HECO) and *Nasella viridula* (NAVI) (Figure 2).

Germination Rate by Soil Type

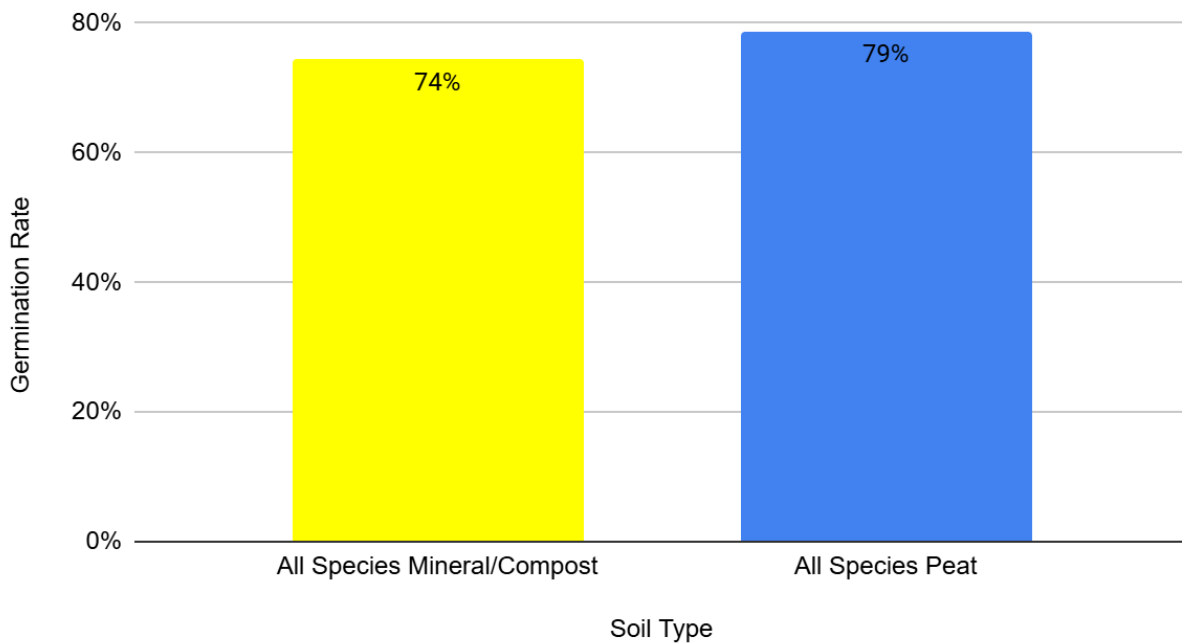


Figure 2. All species germinated in peat had an average germination rate of 79% and all species germinated in mineral compost mix had an average germination rate of 74%.

Germination rates were lowest for *Achillea millifolium* (ACMI) and *Hesperostipa comata* (HECO). *Artemisia ludoviciana* (ARLU), *Heterotheca villosa* (HEVI), and *Symphyotrichum ericoides* (SYER) had the highest germination rates (Figure 3). Overall, the germination rates between peat and MC were very similar.

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Germination Rates by Soil Type

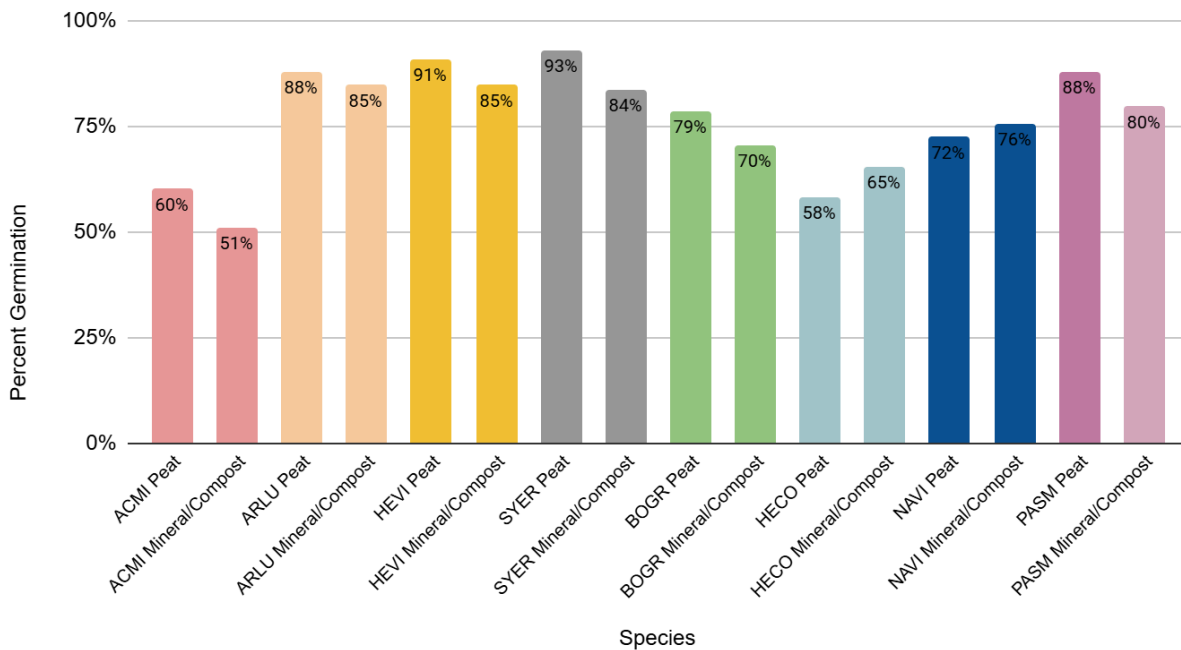


Figure 3. Germination rates per species in each soil type revealed higher germination rates in peat for six of the eight species.

Field Results

Average survival of all species was lowest in the July planting date (48%) and highest in the September planting date (72%). The August planting date had an intermediate average survival rate (62%). (Figure 4).

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Survival of all species by planting date

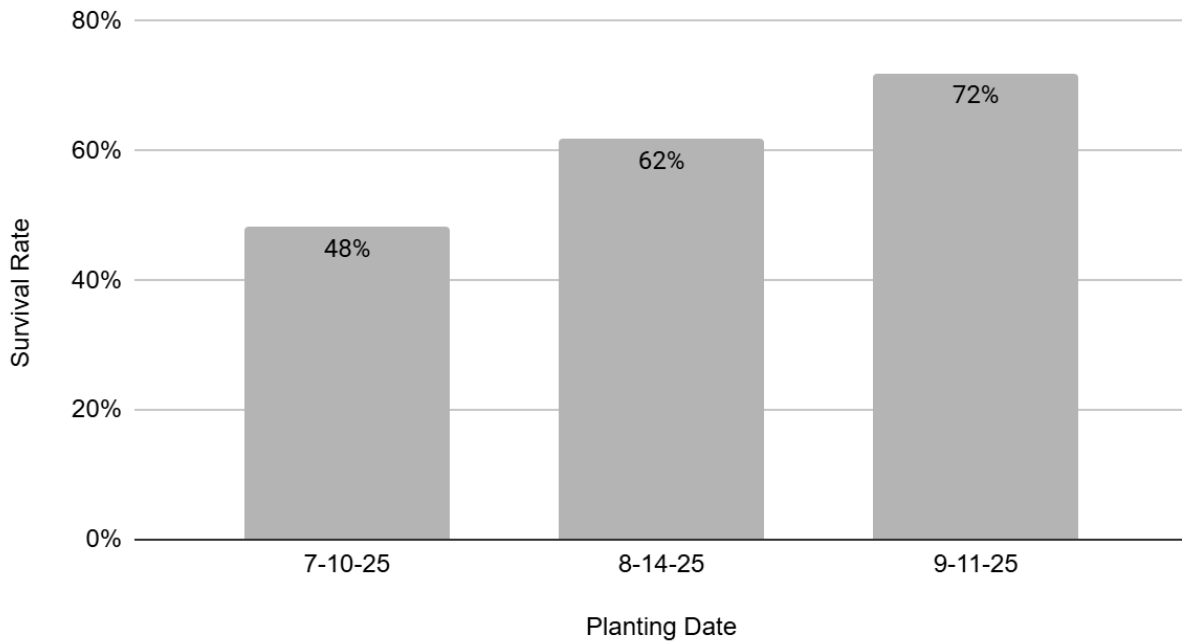


Figure 4. Species were planted on three dates throughout the summer of 2025.

When we look at the data parsed out by planting date and soil type, we see a more complex story. The plants that survived the least were clearly ones planted in the peat and garden soil plots. Plants in the mineral compost mix had impressive survival rates in both July (88%) and September (81%) as compared to average survival rates in peat in July (47%) and in September (74%), and in garden soil in July (46%) and August (62%) (Figure 5).

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Survival of all Species by Soil Type and Planting Date

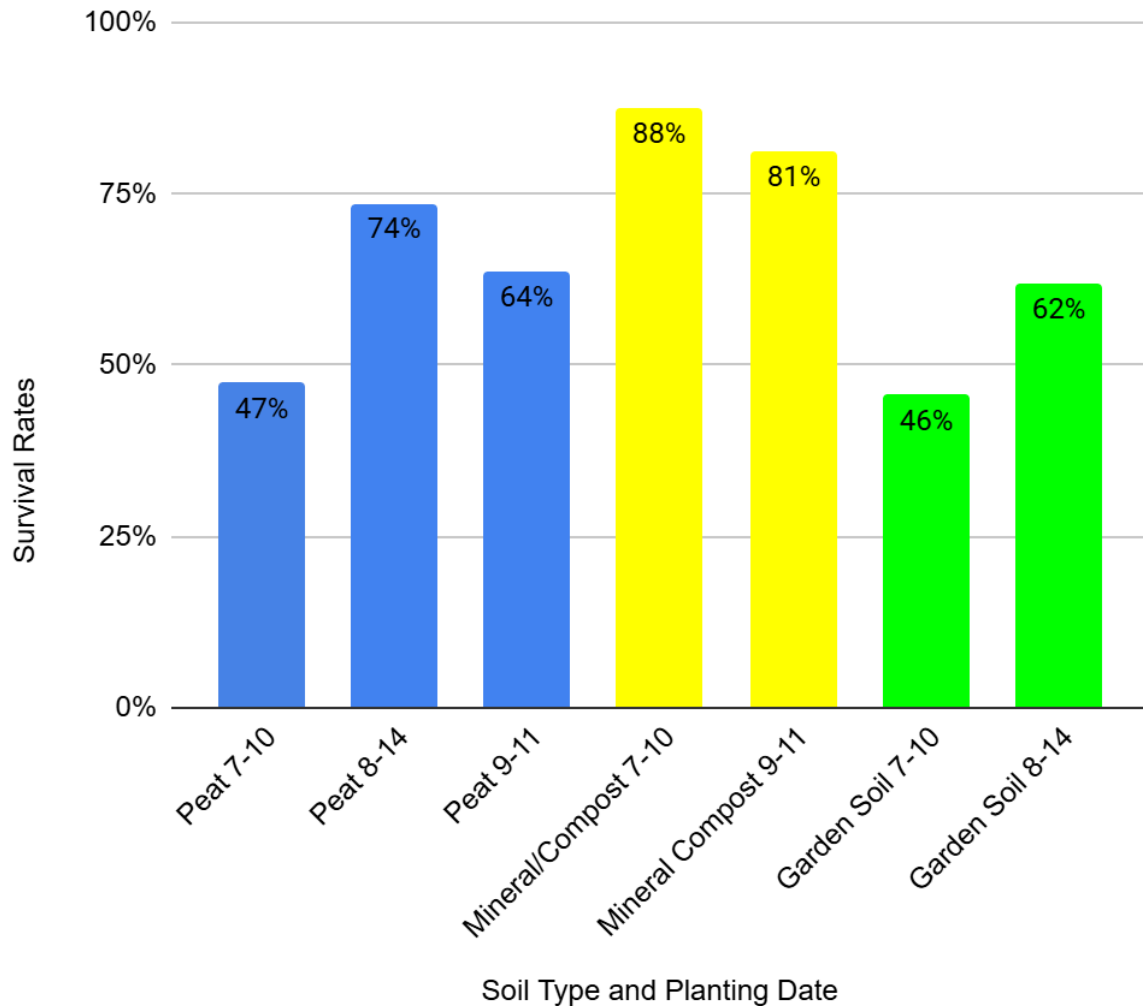


Figure 5. Plants in the mineral compost mix showed opposite average survival rate by planting date than those observed in peat and garden soil.

When we compared the survival of all species on all planting dates to their

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soil types, we found one of the most stunning results of the study. The average survival rate of all species in mineral compost soil was 79% (Figure 6). In peat moss, all species average survival rate was 53%. Native garden soil resulted in an intermediate average survival rate of 71%.

Survival of all species planted on all planting dates by soil type

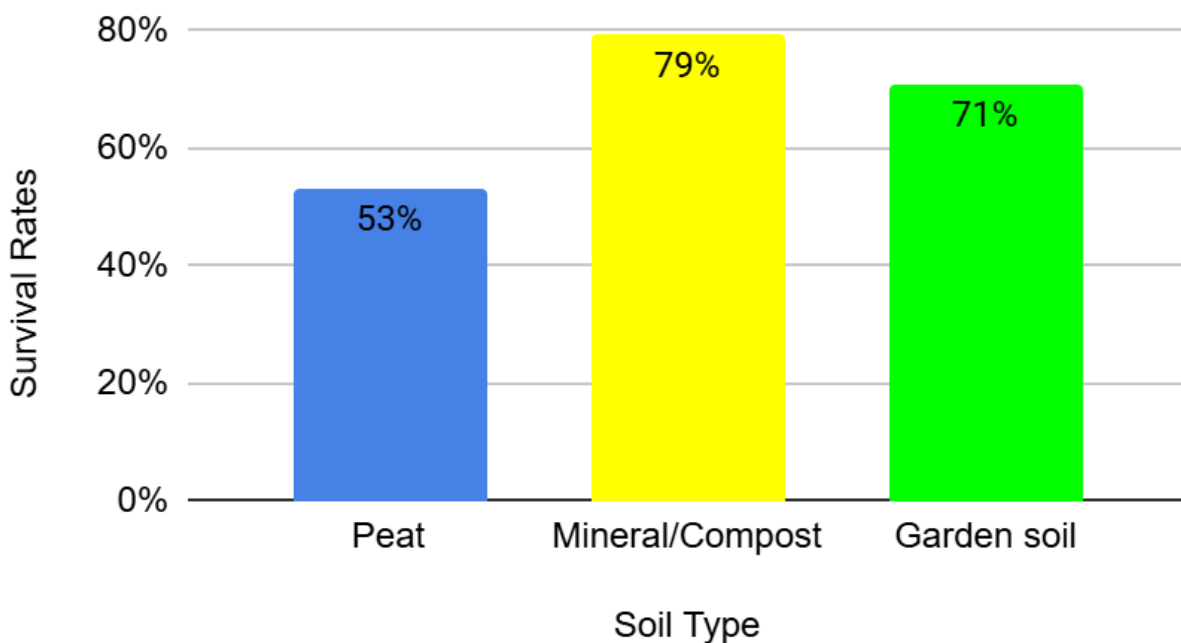


Figure 6. Average survival was highest in the mineral and compost mix.

When we looked at a more nuanced view of individual species by soil type, we saw the same trend repeated per species (Figure 7). The only clear exception was *Artemisia ludoviciana* (ARLU) which had a higher average survival rate in garden soil (78%) compared to 68% in mineral compost mix and 67% in peat.

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Pascopyrum smithii (PASM) had nearly identical average survival rates in mineral compost mix (47%) and in peat (48%). One remarkable result was 100% average survival in *Heterotheca villosa* (HEVI) in mineral compost mix. The most poorly performing species was *Nassella viridula* (NAVI) in peat (11%).

Survival of all Species by Soil Type

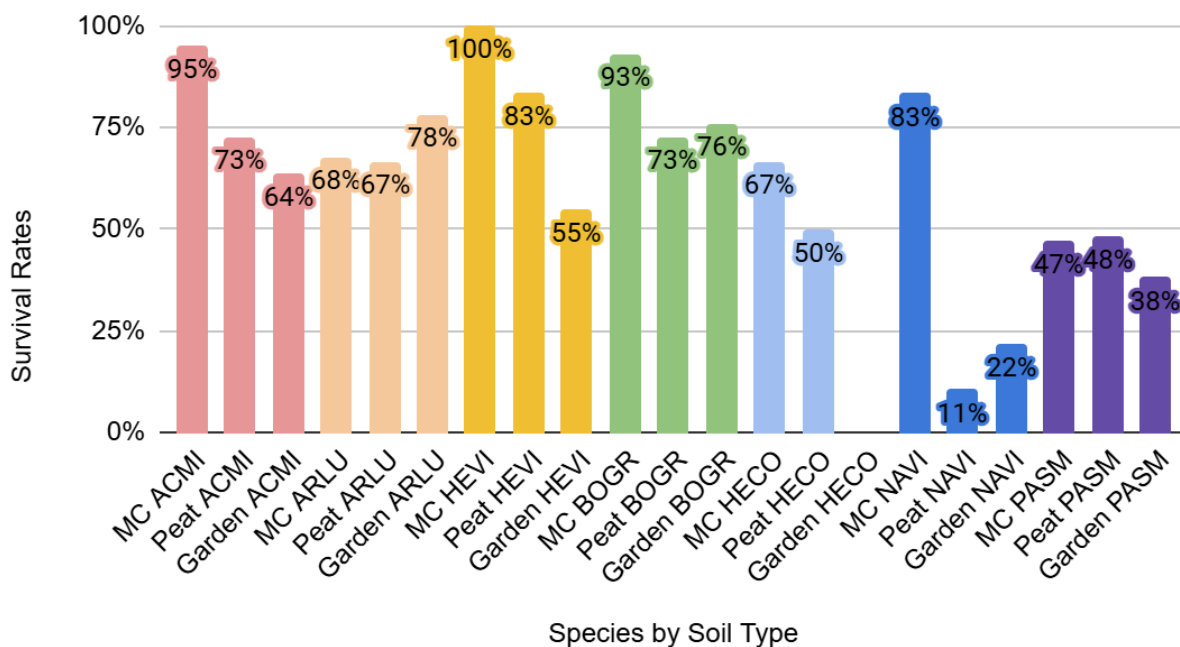


Figure 7. Most species had the greatest average survival rate in mineral compost mix.

When the data is parsed based on fertilization regime, we found that fertilization had a positive effect on all soil treatments (Figure 8). We saw the same

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trend when we looked at fertilization by soil type. In peat, fertilization resulted in a 58% average survival rate compared to 48% in non-fertilized plots. In garden soil, fertilization resulted in 65% average survival rate compared to 42% in unfertilized garden soil. In the mineral compost mix, the effect of fertilization was minimal with 80% survival in fertilized plots and 79% in unfertilized plots (Figure 9).

Survival of All Species by Fertilization Regime

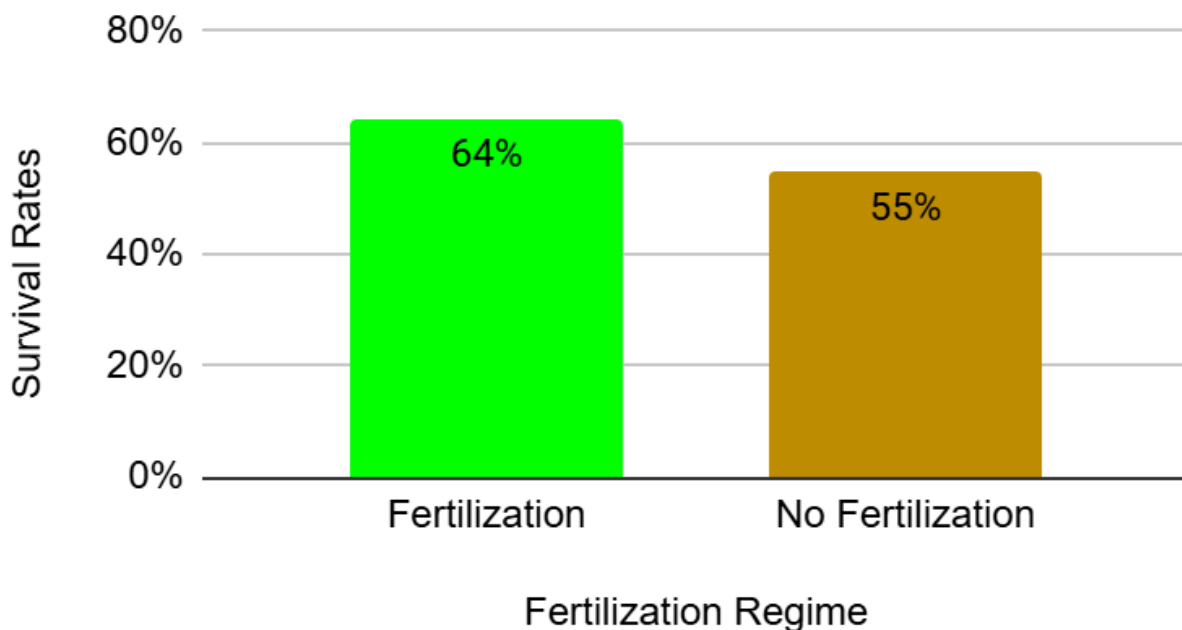


Figure 8. When all species were aggregated by fertilization regime, average survival rates were higher among individuals fertilized throughout the hardening off period.

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Survival of all Species in All Soil Types by Fertilization Regime

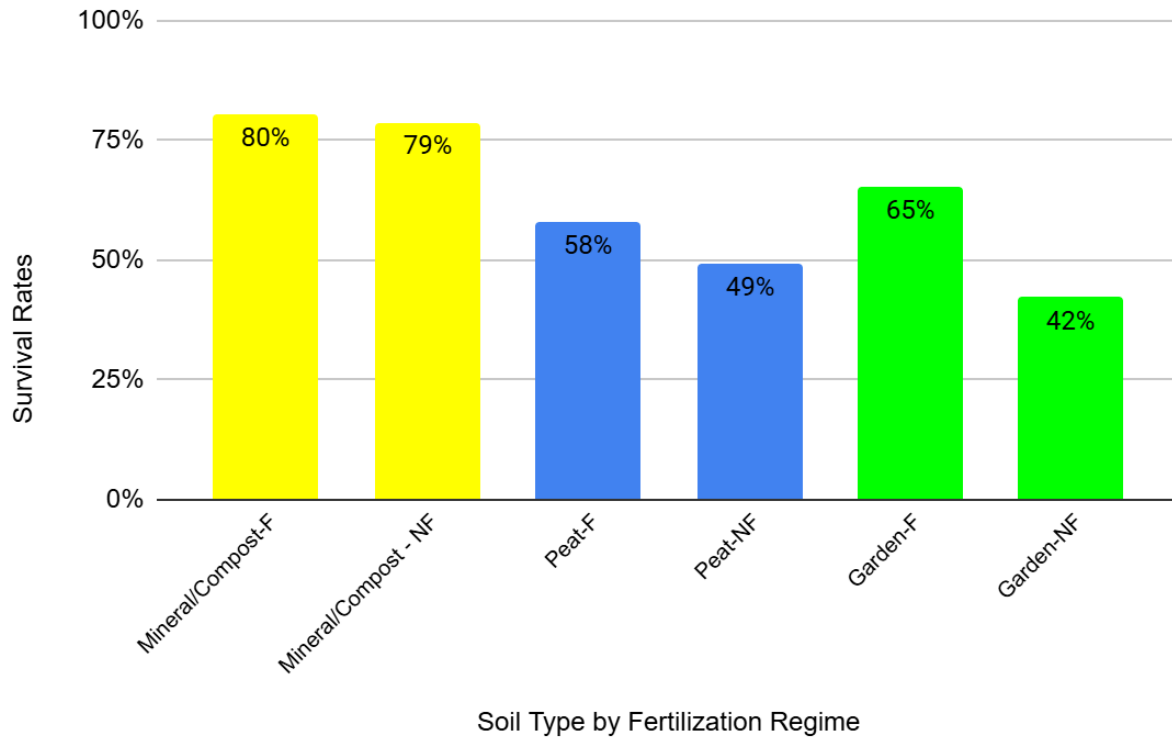


Figure 9. When we compared fertilization to soil type, we found that the positive trend of fertilization was consistent across soil types.

Finally, regardless of planting date, soil type, or fertilization treatments, the species chosen had variable average survival rates (Figure 10). Furthermore, forbs as a group were more successful than grasses (Figure 11). The forbs species had an average survival rate of 70-74%. Grasses had an average survival rate of 28-79%. The great range of survival between grass species and within

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grass species across treatments leads us to question the reliability of our grass data. As noted, it was very challenging to find the grasses, especially with the encroaching non-native *Bromus inermis*.

Survival of all Species

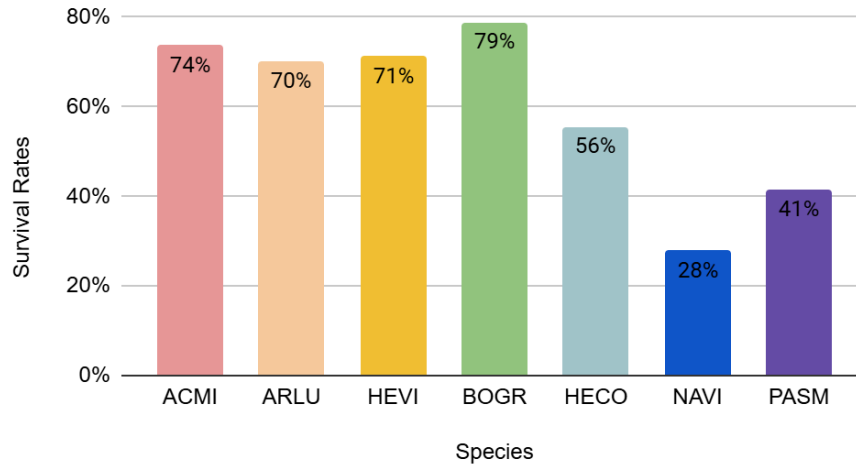


Figure 10. Forb species performed better than grasses, with the exception of *Bouteloua gracilis* which was the top performer in the study.

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Survival of Species by Plant Type

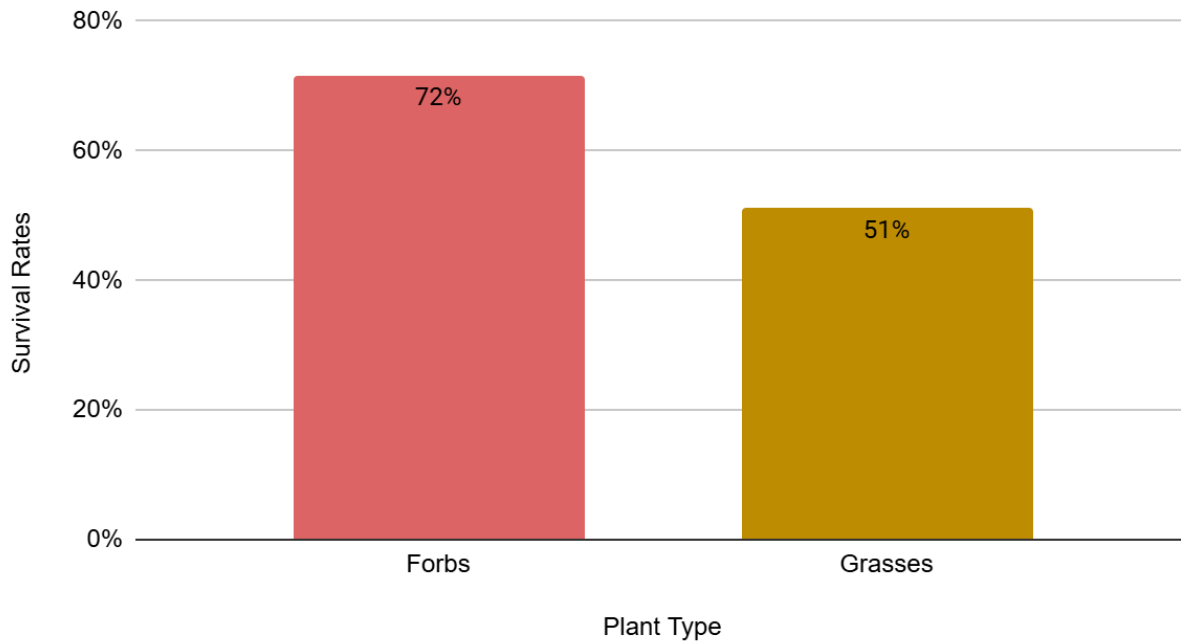


Figure 11. Forbs had an average survival rate of 72% while grasses had an average survival rate of 51%.

Data Analysis

We ran Analysis of Variance (ANOVA) tests in R Studio. ANOVA tests compare variation in survival rate within and across variables. We ran all possible combinations of species, plant date, soil type, and fertilization. Across all the variable combinations, we found only three that showed statistically significant variation across the groups: species comparisons, species and fertilization, and

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soil type and species. We used a Tukey test, or Tukey's Honestly Significant Difference (HSD) test, to determine which pairs of variables have statistically significant differences in their means.

The ANOVA results indicate a significant effect of species on survival ($F = 4.581$, $p = 0.000817$). The variation between the group means (differences between species) is greater than the variation within the groups (natural variation in each species). Based on the results from the Tukey test ACMI, ARLU, BOGR, and HEVI are all statistically similar in terms of survival rate, however they are all statistically different from NAVI.

The ANOVA results indicate a significant effect of species and soil type on survival ($F = 2.481$, $p = 0.00771$). The Tukey test for MC_BOGR is statistically different from peat_NAVI. All other groups' survival rates are not statistically different from any other groups.

The ANOVA results indicate a significant effect of species and fertilization on survival ($F = 4.581$, $p < 0.000817$). Tukey test results suggest BOGR_F, and BOGR_NF are statistically similar in terms of survival rate, however they are all statistically different from NAVI_F. All other groups' survival rates are not statistically different from any other species.

Soil Analysis

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Soil chemical analysis characterized each soil type and compared them to the native soil in the field plots. The average pH was 8.3 for MC, 7.0 for S3, 6.0 for peat, and 7.0 for the field site soil. The average soluble salts (EC) were 0.0 mmho/cm for MC, 0.1 mmho/cm for S3 and 0.01 mmho/cm for peat. Native field plot soils EC was 0.1 mmho/cm. The average organic matter content (OM) for MC was 2.3%, 5.1% for S3, and 4.8% for native field plot soil. The Cation Exchange Capacity (CEC) for MC was 10 meq/100g 16 meq/100g for SC. Native field plot soil CEC was 27.3 meq/100g (Table 2).

Macronutrients measured in MC soil were nitrate (<1 ppm), phosphorus (63 ppm), potassium (310 ppm), calcium (1288 ppm), magnesium (310 ppm), and sulfate-sulfur (3 ppm). Macronutrients measured in S3 were nitrate (9 ppm), phosphorus (92 ppm), potassium (586 ppm), calcium (2394 ppm), magnesium (325 ppm) and sulfate-sulfur (6 ppm). Macronutrients measured and averaged in native field plot soils were nitrate (5.4 ppm), phosphorus (10.6 ppm), potassium (713.5 ppm), calcium (3931.2 ppm), magnesium (678.3 ppm) and sulfate-sulfur (3.1 ppm). We did not measure macronutrients in peat because peat is a soil media that is not yet decomposed. Therefore, the macronutrients are not yet available for plant growth (Table 2).

Micronutrients measured in MC soil were sodium (81 ppm), zinc (4.3 ppm),

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iron (24.7 ppm), and manganese (3.6 ppm). Micronutrients measured in S3 soil were sodium (22 ppm), zinc (47.5 ppm), iron (42.8 ppm), and manganese (6.5ppm). Micronutrients measured and averaged in native field plots were sodium (22.7 ppm), zinc (24.0 ppm), iron (98.1 ppm), and manganese (24.0 ppm) (Table 2).

Table 2. Soil chemical analysis results characterize native soil in the research sites and experimental soil types.

Soil type	pH	EC mmho/cm	OM %	Nitrate-N ppm	P ppm	K ppm	Ca ppm	Mg ppm	Na ppm
field plots averaged	7.0	0.1	4.8	5.4	10.6	713.5	3931.2	678.3	22.7
MC	8.3	0.0	2.3	<1	63	310	1288	310	81
S3	7.2	0.1	5.1	9	92	586	2394	325	22
Peat	6.0	0.01							

Soil Type	CEC meq/100g	S ppm	Zn ppm	Fe ppm	Mn ppm	Cu ppm
Field plots averaged	27.3	3.1	24.0	98.1	24.0	4.2
MC	10	37	4.3	24.7	3.6	1.1
S3	16	6	47.5	42.8	6.5	1.8
Peat			28	3441	74	10

Soil type	% Sand	% Silt	% Clay	Texture by Hydrometer
field sites averaged	59.1	16.0	24.8	Sandy clay loam
MC	75	13	13	Sandy Loam
S3	68	18	14	Sandy Loam

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Spring Monitoring Effort

On April 23, 2026, we returned to the research site to measure plant survival after the winter season. We observed variable phenology across sites. We observed that some species were growing while others appeared to be just emerging or not yet emerged. Therefore, it was impossible to ascertain if the individuals had not survived or had not yet emerged. We measured richness. There were 36 unique species present in the plots at this time (Table 3). Species present in the plots in April 2026 included 6 of 7 species planted, excluding *Hesperostipa comata*.

Table 3. Species present in the plots in April 2026 included 6 of 7 species planted, excluding *Hesperostipa comata*. Planted species are noted with an asterisk.

Scientific Name	Common Name
* <i>Achillea millefolium</i>	common yarrow
<i>Alyssum simplex</i>	field Alison
<i>Andropogon gerardi</i>	big bluestem
<i>Androsace</i>	rock-jasmines
* <i>Artemisia ludoviciana</i>	silver wormwood
* <i>Bouteloua gracilis</i>	blue grama
<i>Bromus inermis</i>	smooth brome
<i>Camelina microcarpa</i>	littlepod false flax
<i>Centaurea stoebe</i>	spotted knapweed
<i>Cirsium arvense</i>	creeping thistle
<i>Cyperaceae</i>	sedges
<i>Descurainia sophia</i>	flixweed

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<i>*Elymus smithii</i>	western wheatgrass
<i>Erigeron canadensis</i>	horseweed
<i>Helianthus pumilus</i>	little sunflower
<i>*Heterotheca villosa</i>	hairy false goldenaster
<i>Hymenothrix dissecta</i>	ragleaf bahia
<i>Lactuca serriola</i>	prickly lettuce
<i>Linaria dalmatica</i>	Balkan toadflax
<i>Lomatium orientale</i>	Oriental desert-parsley
<i>Mertensia lanceolata</i>	prairie bluebells
<i>*Nassella viridula</i>	green needlegrass
<i>Oenothera</i>	evening primrose family
<i>Pedimelum tenuiflorum</i>	slimflower scurfpea
<i>Phacelia heterophylla</i>	virgate scorpionweed
<i>Rosa woodsii</i>	Woods' rose
<i>Sisymbrium altissimum</i>	tall tumbled mustard
<i>Sphaeralcea coccinea</i>	scarlet globemallow
<i>Symphyotrichum falcatum</i>	white prairie aster
<i>Symphyotrichum laeve</i>	smooth blue aster
<i>Taraxacum officinale</i>	common dandelion
<i>Tragopogon dubius</i>	yellow salsify
<i>Verbascum thapsus</i>	great mullein
<i>Vicia</i>	vetches
<i>Yucca glauca</i>	great plains yucca

Discussion

Germination Results

The germination rates were quite high across the study (Figure 2). This is likely due to the highly controlled environment of the Colorado Springs city greenhouse. In some cases, we surmised that some of the ungerminated SC10

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pots were probably not getting misted well. Ideally, we would have rotated the trays but due to their immense weight, this was not possible. Instead, we attempted to change the position of the mist spray slightly each time we measured to ensure that every cell was being adequately watered. In retrospect, we should have set up the trays so they were half peat cells and half mineral compost cells so we could lift them more easily to rotate them on the bench.

When we compared germination rates by species and soil type, we found that germination rates were slightly higher in peat moss (Figure 3). We presume that the increased water holding capacity of peat moss is a possible explanation for this result. Also, it is important to consider the design of a seed. Soil media should not have an effect on germination rates since all biotic needs of the embryo are supplied by the seed environment itself. Abiotic factors like adequate water and warm temperature are signals that break dormancy in the seed. The soil itself is not providing any enrichment until the seed germinates and the plants begin to grow.

We found that the germination rates were not as high in the July planting. We assume that by July, it is too hot in the greenhouse for a conducive seeding environment. Another possibility is that the environmental signalling for the plant is amiss. By July 1, past the summer solstice, the days are already getting shorter, which is typically a signal to plants to start flowering, but not a signal for breaking

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dormancy. Typically, the Colorado Springs horticulture department takes a break from seeding in the hot summer months. Although this is an anecdotal conversation presently, future research might investigate the success of germinating native seeds at different times of year.

Field Results

It is clear that the planting date had an effect on the plant material (Figure 4). After a wet and cool spring, we had a dry and hot midsummer. Plants installed in July had the lowest average survival rates. It was interesting to hear from JCPOS staff that they always plant in September. Currently, it appears that this is the appropriate planting date to encourage survival. However, we are most dubious of the September survival data because we only continued monitoring through October 23, 2025. We look forward to seeing if these results change when we monitor in 2026. We assume that it is difficult for grasses to establish themselves this late in the season and therefore, we expect that the September planting date average survival rate will decrease by 2026.

When we calculate average survival by soil type and planting date, we saw a much more varied picture (Figure 5). Average survival was more varied within the peat planting dates and within the garden soil planting dates. In contrast, the plants in mineral compost showed high survival rates in both July and September, the only months in which they were planted. This result might be due to the large

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plant set for *Bouteloua gracilis*, which had remarkable survival rates as compared to other species. Considering the expansive range of *Bouteloua gracilis*, perhaps this species is more plastic in its habitat and environmental requirements.

When we aggregated these data based on soil type, we found that the plants in the mineral compost mix had the highest average survival rate as compared to the peat and garden soil (Figure 6). This is a trend that repeated itself in all of the permutations of the treatment levels we explored to date. The native soil repeatedly revealed an intermediate average survival rate as compared to the plants in peat or mineral and compost mix. It is probable that we will find that the garden soil had an intermediate soil texture and chemistry as well.

A potentially confounding factor in the comparison of S3 to peat and MC is the different container size used for these plants. We had germinated extra individuals of some species. Rather than terminate these plants, we planted them in S3 and in recycled 3" pots because we did not have any more SC10s. We thought this would be a good "insurance plan", but we never intended to employ the S3 plants in the field. However, during the hardening off period, we lost a significant number of plants. The weather changed rather dramatically in the second hardening off week, from precipitation every day to 100 temperatures and no precipitation. We maintained our hardening process, rather than increasing the regime to respond to the changing weather. Although maintaining the water

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reduction schedule seemed appropriate from a scientific methodology perspective, this was a mistake. We lost plants of all species, except BOGR in MC in one week. We later learned that JCPOS typically waters plants until the day they go to the field. We should have discussed this detail more thoroughly and perhaps not attempted to harden them off but rather, continue a daily watering schedule. We ultimately made the decision to employ the garden soil plants in 3" pots so we could bring an appropriate number of plants to the field. While using garden soil produced interesting results from a horticultural perspective, it complicated the study design and subsequent statistics.

When we compared individual species by soil type, we saw the same trend (Figure 7). We can also appreciate the varied success by species. On the left of the graph are all the forb species and on the right are the grass species. With the exception of *Bouteloua gracilis* (BOGR), the remaining three grass species, *Hesperotipa comata* (HECO), *Nassella viridula* (NAVI), and *Pascopyrum smithii* (PASM) had low survival rates, in general.

We do not feel very confident in these results because it was extremely difficult to locate the young native grasses, especially as the season continued and the plots became repopulated with *Bromus inermis*. It became a bit easier to locate these grasses as they began to senesce into different colors. We found plants in October that we had not been able to find throughout August and

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September. A better system of marking the plants or clearing plots would be a huge improvement for future research. At first, we agreed not to leave plant tags around the site, but every time we had a monitoring event, volunteers devised new strategies to mark and find the plants.

Improving our monitoring skills was one of the key benefits to biweekly monitoring. Ultimately, we analyzed only the data collected at the beginning of the field season and at the end. The beginning measurements were reliable because they were based on the total number of plants planted. The final measurements were strong as well because we had our monitoring protocol dialed down and the grasses were changing colors, making them easier to locate.

Figures 8 and 9 show the positive effect of fertilization. There are many explanations for this, not the least of which being the lean nature of native soils. Fertilization perhaps strengthened the plants enough to aid their drought tolerance in the field. We looked at the root growth in all treatments before planting out. There was no discernable difference in the root development across treatments, nor were there any apparent visual differences in the above ground plant parts between the fertilized and unfertilized sets. Despite a lack of visual cues, clearly fertilization is a strength to retain in future plant growing initiatives. A future study might compare fertilization techniques and chemicals. In this study we used an organic nitrogen-based fertilizer applied weekly with a mist pump sprayer.

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It is worth noting that the species did not perform equally when compared to each other. The species used were chosen by JCPOS staff. They were certainly appropriate to the field site, as we have seen all of the species used around North Table Mountain Park. However, from a practice orientation, it would be efficient and effective to ascertain in a future study what species result in the greatest survival potential. The top performers were all the forbs, *Achillea millifolium* (ACMI), *Artemisia ludoviciana* (ARLU), and *Heterotheca villosa* (HEVI), and *Bouteloua gracilis* (BOGR). The other three grasses were more challenging opportunities, *Hesperostipa comata* (HECO), *Nassella viridula* (NAVI), and *Pascopyrum smithii* (PASM).

This was an interesting result because NAVI and PASM are widely used in a restoration context already. We think we did not seed the pots heavily enough. This might have helped us see the plants better in the field and also might have given them a fighting chance for survival because there would be more individuals in the pot. It was so easy to see BOGR because of its growth form. It proliferated many leaves while the other species tended to increase in height but maintain only a few leaves. It was easy to see the BOGR as the season progressed because the leaves curled as they senesced. It doesn't seem likely that NAVI would be so successful in mineral compost mix and dramatically unsuccessful in peat (Figure 7) and therefore, we question this result.

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The statistical analysis revealed, with few exceptions, that the species did not behave differently from each other under treatment. This is a positive outcome, as it suggests that the species selection did not play a significant role in survival rates. It also suggests that there was not a significant effect of treatment on plant survival. BOGR and NAVI were statistically different from each other and NAVI was statistically different from all other species. However, we are not confident in these results due to their discrepant sample sizes.

We attribute the lack of statistical significance to the number of variables employed. In a future study, we would restrict the variables under investigation. Reducing the variables under investigation would result in larger sample sizes, which could uncover the true effect of our treatments.

One variable that we did not consider before starting the effort was the variable skill of volunteers in planting the material. We could not measure this source of error, but immediately observed it. Some plants were clearly planted more deeply than others. Plants were also packed into the soil with variable effectiveness. It was clear that some plants were popped out of the ground when we returned for monitoring. Others exhibited herbivory. We thought that perhaps that as wildlife was consuming the plant, it just came out of the ground. In some cases, we could see the plant nearby. In others, the plant was simply gone. In a

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future study, we would conduct a separate training to teach volunteers how to plant in the field. This could reduce the error arising from this planting variable.

Soil Analysis

We found that MC had a higher pH than any other soil type. This could be due to the mineral component used for the MC soil. It was harvested from a quarry in El Paso County, not from a Jefferson County quarry. Perhaps there are significant differences in the base soil types. We were surprised to find the average pH for the field sites was 7.0. Colorado has basic soils typically. We can surmise that the plants would prefer the native and neutral soils and thus would expand their roots behind the planting medium when they were planted in the restoration site.

We found EC was extremely low in all soil types. This is favorable for plant growth, but somewhat surprising giving the typically high EC of Colorado native soils caused by mineral weathering.

Organic matter content (OM) in S3 and native field plot soils was similar (5.1% and 4.8% respectively. Organic matter content in MC soil was low (2.3%). Organic matter helps the soil retain moisture. The very low OM in MC soil may have contributed to our hardening off losses in MC soil before bringing the plants to the field. Although we expect low OM in Colorado native soils, it may be

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facultative to increase OM in MC mix to increase water holding capacity in the restoration context.

Lime is often used to adjust soil pH. High lime content is a measure of how resistant the soil is to changes in pH. In the restoration context, soil amendments are not applied and therefore these results were expected. Furthermore, the reported pH levels were close to neutral and therefore appropriate for plant growth without any soil amendment.

The Cation Exchange Capacity (CEC) is a measure to indicate the soil's ability to hold nutrients. This measure was low for MC (10 meq/100g) and S3 (16 meq/100g) and much higher for native field soil (27.3 meq/100g). This indicates that native soils have a greater capacity to hold nutrients than the experimental soils. This suggests that the native soil is more facultative to plant growth than the experimental soil, which in turn, encourages root expansion into the field site soils.

Nitrate was very low in MC (<1 ppm), low in native field plot soil (5.4 ppm), and medium in S3 (9 ppm). Colorado soils are typically low in nitrogen because there is little organic matter from which nitrogen species may decompose and become plant available. Therefore, these results are expected. These low values may help explain why there was an effect from fertilizer application before field planting. Higher nitrate value in S3 is possibly a result of the compost used in the

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S3, which is derived from chicken waste, which typically exhibits higher nitrogen than non-animal derived compost materials, like that used in MC.

Phosphorus was very high in MC (63ppm) and S3 (92 ppm) and optimal in native field plot soils (5.4ppm). Potassium was very high in all soil types (MC = 310 ppm, S3 = 586ppm, native field plot soils (713.5ppm). Potassium was very high in MC (310 ppm), in S3 (586 ppm), and in native field plot soils (713.5 ppm). Calcium was optimum in MC (1288 ppm) and S3 (2394 ppm), and high in native field plot soils (3931.2 ppm). Magnesium was very high in all soil types (MC = 310 ppm, S3 = 325 ppm, native field plot soils = 678.3 ppm). Sulfate was low in MC (3 ppm), medium in S3 (6 ppm), and low in native field plot soils (3.1 ppm).

Low values in macronutrients and micronutrients can inhibit plant growth, but we did not observe this outcome except in the case of sulfate. We therefore conclude that the soils are not inhibiting plant growth. Given that we did not continue to fertilize once the plants were in the field, we can conclude that these values are a result of the basement materials in North Table Mountain Park soils. We can also conclude that our experimental soils were less facultative than native field plot soils. The lean design of MC soil is intended to encourage plants to expand their root systems when they enter the field. We did not find that the experimental soils were more facultative than the native soils and therefore we conclude the root expansion is likely into native soils where they were planted. A

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future study could measure this by terminating the plants at the end of the study to measure total biomass of plants. This is usually accomplished inside of the controlled greenhouse environment where plants can be removed from containers without damaging their roots to ascertain in which soil type the plants grew the most.

Sodium levels were highest in the MC soil type (81 ppm), while comparable between S3 and the native field plots (S3= 22.0 ppm, native field plot soils= 24.0 ppm). Zinc levels were very high in the MC soil type, S3 soil type, and native field soil average (MC= 4.3ppm, S3= 22 ppm, native field plot soils= 24.0 ppm) Any value >2.0 ppm Zn is considered “very high.” Iron levels in the MC soil were high (24.7 ppm), very high in the S3 soil type (42.8 ppm) and the highest in the native field plot soils (98.1 ppm). A “very high” amount of iron in a soil is any value above 30 ppm. Manganese amounts were lowest in the MC soil type (3.6 ppm) but considered to be optimum. Manganese levels in the S3 soil type were high (6.5 ppm) and in the native field plots very high (24.0 ppm). Any value >10 ppm is considered very high in manganese.

Spring Monitoring Effort

We found that our 2025 planting season measurement techniques were not suitable for the spring monitoring. In 2025, numerous individuals of the same

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species were planted adjacent to each other. Then we measured the presence of each individual to calculate our percent survival results. By spring 2026, it was impossible to discern individual plants of some species already growing. Some plants were expanded into the whole area where the species was planted.

Moving forward, adopting a percent cover calculation is appropriate for established plants. In a community science context, it might be appropriate to use a fixed photo point that is measured digitally to calculate percent cover. The photo point could describe the entire plot. Alternatively, a randomly placed quadrat could be measured for richness, percent cover, number of flowering stems, etc.

At the time of the spring monitoring effort, while some species were actively growing, other species had not yet emerged, or were just barely getting started. Another monitoring effort later in the season could have been helpful, but new volunteer programming that alters the site conditions is now engaged in the research site.

One fascinating observation was that the phenology was very different from west to east across the plot matrix in the site. Plots 1-6, the more westerly plots, had phenological characteristics far more advanced than plots 7-12, in the more easterly rows of plots where the vegetation was still largely dormant. This effect of aspect was very interesting to observe. Upon departing the site, we showed the

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volunteers the original site location, near to a future parking lot expansion. We remarked that in that site, aspect would not have played such a strong role in the plant response.

We also observed variable richness in the plots. This could be due to the phonological differences, but perhaps aspect plays a bigger role than expected. We also observed variable richness throughout the 2025 field season. While the plants we planted were the same in each plot, the plants that grew in the plots that we did not plant were variable. For example, there was a lot of *Rosa woodsii* and *Cirsium arvense* in more easterly plots.

The only planted species we did not observe in April 2026 was *Hesperostipa comata*. Very few of *H. comata* made it to the field in the 2025 planting season so this is not surprising. Furthermore, this was the most difficult grass to find. The *Pascopyrum smithii* has very apparent veins in the leaves and *Bouteloua gracilis* makes many somewhat curly leaves that appear as clumps as a young plant. *H. comata* and *Nasella viridula* individuals measured in the 2025 field season were single blades. Among the *Bromus inermis*, this was a great challenge. It was easier to see in the fall because grasses senesce to different colors. If we had more heavily seeded the SC10s to establish true plugs, the monitoring effort may be more manageable. It was very easy to see the forb species and thus they were excellent species selections for the community

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science application. Images of planted species and plots can be observed in the [Native Plant Growth Study iNaturalist project](#) and in our [presentation slides](#). We presented this study at the Landscaping with Colorado Native Plants Conference in February 2026. In October 2026, Alex Crochet, Maggie Gaddis, and Anna Theodorakos will facilitate an afternoon workshop about the study, including a native plant propagation training, at the Colorado Open Space Alliance conference. We have also been asked to present at the Front Range Open Space Research Symposium in November 2026.

Conclusion

The results of this study will improve native plant availability for restoration and horticultural applications in JCPOS. The results of this study have the potential to reduce operating budgets by reducing supply and labor costs. Both phases of this research engaged the public in community science initiatives, thereby increasing their 1) awareness of local conservation efforts, 2) botanical knowledge, 3) and monitoring skills. This project has the potential to reinforce and expand JCPOS's existing community engagement initiatives.

It is clear that mineral compost mix was a viable growing medium, if not the preferred media for native plants. In all cross comparisons, the results were the same: average survival rates were higher in the mineral compost mix, lowest in peat, and intermediate in garden soil. Fertilization supported higher average survival rates. Later planting dates also supported the current average survival rates. As we continue to

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analyze the data, we hope to reveal more details that could support JCPOS restoration initiatives.

This line of research comparing native plant growth success in peat and non-peat media has impressive implications for climate resiliency in today's world. No amount of gardening or restoration will restore the carbon sink that is lost when peat is harvested. Furthermore, peat is harvested from thousands of miles away in the boreal forests of Canada. The heavy machinery, fuels, and transportation miles required to get peat to Colorado is also a losing carbon equation. All of these activities are a carbon source, contributing to our climate change concerns. If we start with this heavy burden of carbon sourcing, our restoration efforts will never achieve carbon neutrality, let alone carbon storage. Let's stop the bleeding! Let's grow plants in the soil to which they are adapted. Colorado soils are pedasols, rock-based soils. Peat soils don't develop until you reach the 100th meridian in the midwest of the United States, so why are we trying to grow Colorado native plants in peat? This research is only a pilot study but hopefully it has set the course for JCPOS to take the most climate resilient pathways in its restoration pursuits.

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